

Mathematical Logic (Propositional Calculus)

(For students of Discrete Mathematics/Discrete Structure)

Introduction: The rules of logic give precise meaning to mathematical statements. These rules are used to distinguish between valid and invalid mathematical arguments. Because a major goal of this topic is to teach the students how to understand and how to construct correct mathematical arguments, we begin our study of discrete mathematics/discrete structure with an introduction to logic. Besides the importance of logic in understanding mathematical reasoning, logic has numerous applications to computer science. These rules are used in the design of computer circuits, the construction of computer programs, the verification of the correctness of programs, and in many other ways. Furthermore, software systems have been developed for constructing some, but not all, types of proofs automatically.

Proposition: A proposition is a statement which is either true or false. It is declarative sentence.

Examples: The following statements are all propositions.

- (i) It rained yesterday.
- (ii) If x is an integer, then x^2 is a positive integer.
- (iii) Karachi is the capital of Pakistan.
- (iv) Karachi is the capital of Sindh.
- (v) Islamabad is the capital of Pakistan.

Examples: The following statements are not propositions.

- (i) What is your name?
- (ii) $x^2 = 13$
- (iii) Please report at 11 am Sharp.

Proportional Variables: The lower case letters starting from p onwards are used to represent propositions for example $p, q, r, s \dots$

- (i) $p: 2 + 2 = 4$ (ii) $q: \text{It rained Yesterday.}$

Combination of Propositions: We can combine the proposition to produce new propositions. These are called three fundamental and three derived Connectors to combine the propositions.

A) Fundamental Connectors

1. Conjunction (AND): It means ANDing of two statements. Assume p and q be two propositions. Conjunction of p and q to be a proposition which is true when both p and q are true, otherwise false. It is denoted by $p \wedge q$

Truth table:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

2. Disjunction (OR): It means ORing of two statements. Assume p and q be two propositions. Disjunction of p and q to be a proposition which is true when either one or both p and q are true and is false when both p and q are false. It is denoted by $p \vee q$

Truth table

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

3. Negation (-): It means opposite of original Statement. Assume p and q be two propositions. Negation of p to be proposition which is true when p is false and is false when p is true. It is denoted by $\neg p$

Truth table

p	$\neg p$
T	F
F	T

Examples:

Q1. Show that $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$

Solution

p	q	r	$(q \vee r)$	$p \wedge (q \vee r)$	$p \wedge q$	$p \wedge r$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	T	F	T
T	F	T	T	T	F	T	T
T	F	F	F	F	F	F	F
F	T	T	F	F	F	F	F
F	T	F	F	F	F	F	F
F	F	T	F	F	F	F	F
F	F	F	F	F	F	F	F

Q2. Consider the following.

p : He is rich,

q : He is generous.

Write the proposition which combines the proposition p and q Using Conjunction ($\wedge$), disjunction ($\vee$) and negation ($\neg$).

Solution:

Conjunction: He is rich and generous that is $p \wedge q$

Disjunction: He is rich or generous that is $p \vee q$

Negation: He is not rich that is $(\neg p)$

He not generous that is $(\neg q)$

It is false that he is rich or generous that is $\neg(p \vee q)$

It is false that he is not rich $\neg(\neg p)$

He is neither rich nor generous that is $(\neg p \wedge \neg q)$

B) Derived Connectors

1. **NAND:** It means negation after ANDing of two statements. Assume p and q be two propositions. Nanding of p and q to be a proposition which is false when both p and q are true, otherwise true. It is denoted by $p \uparrow q$

Truth table

p	q	$p \uparrow q$
T	T	F
T	F	T
F	T	T
F	F	T

2. **NOR:** It means negation after ORing of two statements. Assume p and q be two propositions. NORing of p and q to be a proposition which is true when both p and q are false, otherwise false. It is denoted by $p \downarrow q$

Truth table

p	q	$p \downarrow q$
T	T	F
T	F	F
F	T	F
F	F	T

3. **XOR** (Exclusive or): Assume p and q be two propositions. XORing of p and q is true or if p is true or if q is true but not both and vice versa. It is denoted by $p \oplus q$

Truth table

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Examples

Q1. Generate the truth table for $p \oplus q \oplus r$

Solution

p	q	r	$p \oplus q$	$p \oplus q \oplus r$
T	T	T	F	T
T	T	F	F	F
T	F	T	T	F
T	F	F	T	T
F	T	T	T	F
F	T	F	T	T
F	F	T	F	T
F	F	F	F	F

Q2. Generate the truth table $p \uparrow q \uparrow r$

Solution

p	q	r	$p \uparrow q$	$p \uparrow q \uparrow r$
T	T	T	F	T
T	T	F	F	T
T	F	T	T	F
T	F	F	T	T
F	T	T	T	F
F	T	F	T	T
F	F	T	T	F
F	F	F	T	T

Q3. Show that $p \oplus q = (p \wedge \neg q) \vee (\neg p \wedge q)$

Solution

p	q	$p \oplus q$	$\neg q$	$\neg p$	$p \wedge \neg q$	$\neg p \wedge q$	$(p \wedge \neg q) \vee (\neg p \wedge q)$
T	T	F	F	F	F	F	F
T	F	T	T	F	T	F	T
F	T	T	F	T	F	T	T
F	F	F	T	T	F	F	F

Some other Connectors:

1. Conditional: Conditional Statement of the form "If p then q " are called Conditional statements. It is denoted $p \rightarrow q$ and read as " p implies q ". Conditional Statements is true if both p and q are true or if p is false. It is false if p is true and q is false. The proposition p is called hypothesis and q is called conclusion.

Truth table

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Variation in conditional statement:

Contra positive: The proposition $\neg q \rightarrow \neg p$ is called Contrapositive of $p \rightarrow q$.

Converse: The proposition $q \rightarrow p$ is called Converse of $p \rightarrow q$.

Inverse: The proposition $\neg p \rightarrow \neg q$ is called the inverse of $p \rightarrow q$.

2. Bi Conditional: Biconditional Statement of the form "If p and only if" are called biconditional statements. It is denoted as $p \leftrightarrow q$ and read as "If p and only if q " the proposition $p \leftrightarrow q$ is true if p and q have the same truth values and is false if p and q do not have the same truth values. The name of biconditional comes from the fact that $p \leftrightarrow q$ is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$

Truth table

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example

Q1. Show that $p \leftrightarrow q = (p \rightarrow q) \wedge (q \rightarrow p)$

Solution

p	q	$p \leftrightarrow q$	p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T	T	T
T	F	F	T	F	F	T	F
F	T	F	F	T	T	F	F
F	F	T	F	F	T	T	T

Principle of Duality: Two statements A_1 and A_2 are said to be dual of each other if either one can be obtained from the other by replacing $\wedge$ (AND) by $\vee$ (OR) and $\vee$ (OR) by $\wedge$ (AND). Also if formula contain T (true) or F (false). Then by replacing T by F and F by T to obtain the dual. Similarly $\uparrow$ (NAND) and $\downarrow$ (NOR) one dual of each other.

Tautologies: A Proposition p is tautology if it is true under all circumstances. It means it contain only T in the final column of its truth table

Example

Q1. Show that the statement $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$ is a tautology

Solution

p	q	$p \leftrightarrow q$	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$	$(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$
T	T	T	F	F	T	T
T	F	F	T	F	F	T
F	T	F	F	T	T	T
F	F	T	T	T	T	T

Contradiction: A statement that is always false is called a contradiction

Example

Q1. Show that $p \wedge \neg p$ is a contradiction

Solution

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Contingency: A statement that can be either true or false depending on the truth table values of its variables is called a contingency

Q1. Show that the statement $(p \rightarrow q) \rightarrow (p \wedge q)$ is a contingency.

Solution

p	q	$p \leftrightarrow q$	$p \wedge q$	$(p \rightarrow q) \rightarrow (p \wedge q)$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	F
F	F	T	F	F

Equivalence of proposition: Two proposition are said to be logically equivalent if they have exactly the same truth values under all circumstances. The following are fundamental logical equivalent expressions.

1	De Morgan's Law	$\neg(p \wedge q) = \neg p \vee \neg q$
		$\neg(p \vee q) = \neg p \wedge \neg q$
2	Commutative properties	$(p \vee q) = (q \vee p)$
		$(p \wedge q) = (q \wedge p)$
3	Associative properties	$(p \vee q) \vee r = p \vee (q \vee r)$
		$(p \wedge q) \wedge r = p \wedge (q \wedge r)$
4	Distributive properties	$p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$
		$p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$
5	Complement property	$p = \neg(\neg p)$
6	Transposition	$(p \rightarrow q) = (\neg q \rightarrow \neg p)$
7	Implication	$(p \rightarrow q) = (\neg p \vee q)$
		$(p \leftrightarrow q) = (p \rightarrow q) \wedge (q \rightarrow p)$
		$(p \leftrightarrow q) = (p \rightarrow q) \vee (\neg p \wedge \neg q)$
8	Exportation	$(p \wedge q) \rightarrow r = [p \rightarrow (q \rightarrow r)]$

Application of Propositional Logic:

- Translating English Sentences
- System Specification
- Boolean Structure
- Logic Puzzles
- Logic Circuit (Digital Circuit)

Exercise (for home work)

Q1. Let p be "It is hot day" and q be "The temp is 45C" write in Simple sentence the meaning of following.

- (i) $\neg p$
- (ii) $\neg(p \vee q)$
- (iii) $\neg(p \wedge q)$
- (iv) $\neg(\neg q)$
- (v) $(p \wedge q)$
- (vi) $(\neg p \wedge \neg q)$
- (vii) $\neg(\neg p \vee \neg q)$

Q2. Show that $(p \oplus q) \vee (p \downarrow q) = p \uparrow q$

Q3. Show that $(p \uparrow q) \oplus (p \uparrow q) = (p \vee q) \wedge (p \downarrow q)$

Q4. Show that $(p \rightarrow q) = \neg q \rightarrow \neg p$

Q5. Show that $(p \rightarrow q) = \neg(p \wedge \neg q) = \neg p \vee q = \neg q \rightarrow \neg p$

Q6. To obtain the dual $p \wedge (q \wedge r) = p \vee (q \vee r)$

Q7. To obtain the dual $\neg p \vee \neg q = \neg p \wedge \neg q$

Q8. $(p \uparrow q) \oplus (p \uparrow q)$ is a contradiction?

Q9. $(p \vee q) \wedge (p \downarrow q)$ is a contradiction?

Q10. From the following formulas find out tautology, contingency and contradiction

- (i) $p \rightarrow p \wedge (p \vee q)$
- (ii) $(p \wedge \neg q) \vee (\neg p \wedge q)$
- (iii) $\neg(p \vee q) \vee (\neg p \vee q)$
- (iv) $p \vee \neg(p \wedge q)$
- (v) $p \rightarrow (q \rightarrow p)$

George Boole (born November 2, 1815, Lincoln, Lincolnshire, England, died December 8, 1864, County Cork, Ireland), English mathematician who helped establish modern symbolic logic and whose algebra of logic, now called Boolean algebra, is basic to the design of digital computer circuits.

